How-To Library

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Frequently Asked Questions for Cloud Data Integration Kafka Connector

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Abstract

This article describes frequently asked questions about using Kafka Connector to read data from and write to Kafka topics.

Supported Versions

- Cloud Data Integration Kafka Connector Spring 2020 April

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General Questions

What is Apache Kafka?

Apache Kafka is an open-source scalable, fault-tolerant, and distributed data streaming platform. You can process large amounts of data quickly by using Kafka. Kafka can serve as an interim staging area for data consumed by various types of downstream consumer applications.

Kafka runs as a cluster comprised of one or more servers, each of which is called a broker. Kafka brokers stream data in the form of messages that are published to a topic.

Kafka topics are divided into partitions which gives better throughput and scales the number of messages processed. Message ordering is guaranteed only within partitions.

How can I configure a Kafka cluster?

To configure a Kafka cluster, you must have at least three nodes or Kafka brokers. You must create configuration files for each of the broker and start the nodes on ZooKeeper. For more information about configuring a Kafka cluster, see the Kafka documentation.

Which is the latest available version of Cloud Data Integration Kafka Connector?

Kafka Connector is available in the Cloud Data Integration Spring 2020 April release.

Which license do I need to use Cloud Data Integration Kafka Connector?

Kafka Connector is licensed for each Kafka cluster. To use Cloud Data Integration Kafka Connector, you need to enable the Kafka Connector license for your organization.

How do I install Cloud Data Integration Kafka Connector?

Kafka Connector is available as an Add-on Connector in Informatica Intelligent Cloud Services. You can install a free trial version of the Kafka connector or you can buy the connector from Informatica.

What does Kafka Connector use to access a Kafka cluster?

Kafka Connector uses the Kafka Consumer and Producer API version 0.11.0.0 to connect to a Kafka cluster.

What Kafka broker version does Kafka Connector use?

Kafka Connector uses the Kafka messaging broker version Apache 0.10.1.1 and above.

What are the variants of Kafka that Kafka Connector support?

Yes. Cloud Data Integration Kafka Connector supports the following variants of Kafka clusters:

- Apache Kafka
- Confluent Kafka

Apache Kafka or Confluent Kafka clusters can run locally or on Cloud.

What information do I need to create a Kafka connection?

When you create a Kafka connection, you must have the following information:

Kafka Broker List

The IP address and port combinations of the Kafka messaging system broker list. You can enter multiple comma-separated IP address and port combinations.

Retry Timeout in seconds

Number of seconds the Secure Agent attempts to reconnect to the Kafka broker to write data.

Kafka Broker Version

The Kafka messaging broker version. Kafka Connector supports **Apache 0.10.1.1 and above** versions of the Kafka broker.

Additional Connection Properties

Comma-separated list of additional configuration properties to connect to the Kafka broker in a secured way through a Kerberised Kafka cluster or connect to a Kafka broker over SSL.

SSL Mode

Determines whether the Secure Agent establishes a secure connection to the Kafka broker. You can select disabled, require, one-way, or two-way as the SSL mode.

SSL TrustStore File Path

The complete path and file name of the truststore file. The truststore file contains the SSL certificate that the Kafka cluster validates against the Kafka broker certificate.

SSL TrustStore Password

The password for the truststore file

SSL KeyStore File Path

The complete path and name of the Java keystore file. The keystore file contains the certificate that the Kafka broker validates against the Kafka cluster certificate.

SSL KeyStore Password

The password for the keystore file.

Additional Security Properties

Comma-separated list of additional configuration properties to connect to the Kafka broker in a secured way through a Kerberised Kafka cluster or connect to a Kafka broker over SSL in encrypted format. The Additional Security Properties overrides the Additional Connection Properties.

For more information about creating a Kafka connection, see the *Cloud Data Integration Kafka Connector documentation*.

What are the operations that Cloud Data Integration supports for Kafka Connector?

You can use Kafka Connector to read from and insert data to Kafka topics.

What are the file formats supported by Kafka Connector?

Kafka Connector supports the following file formats:

- Binary
- JSON
- Avro

When you import a Kafka topic, you can specify the format in which the Kafka Connector reads data from or writes data to a Kafka topic. You can specify JSON or Avro as format.

When you specify JSON format, you must provide a sample JSON schema file.

The following schema is a sample JSON schema:

```
{"id" : 1, "name" : "sample"}
```

Note: The JSON schema must be defined in a single line. If you define a JSON schema that contains more than one line, the mapping fails.

When you specify Avro format, provide a sample Avro schema as a `.avsc` file.

The following schema is a sample Avro schema:

```
{
  "type" : "record",
  "name" : "Tree",
  "fields" : [
{"name" : "children", "type" : ("type" : "array", "items": "Tree"}}
]}
```

Does Kafka Connector support the Confluent Avro format or Schema Registry?

No. Currently, Kafka Connector does not support the Confluent Avro format or Schema Registry.

Does Kafka Connector supports commit interval?

No. Commit interval is not applicable when you configure a Kafka mapping.

What is the default file format when I import a Kafka topic?

When you import a Kafka topic and do not select a format type, Kafka Connector imports the data from the Kafka topic in binary format.

How can I configure Topic Pattern to read data from Kafka topics in a Cloud Data Integration mapping?

When you configure a Kafka mapping to read data from a Kafka topic, you can specify a regular expression for the topic name pattern in the **Topic Pattern** advanced source property.

How do I access the sample data or schema of a Kafka topic?

You can preview the Kafka source or target data when you import a Kafka topic.

How can I configure a secured connection to Kafka?

Kafka Connector supports one-way SSL, two-way SSL, and Kerberos Authentication protocols. You can configure SSL specific properties in the Kafka connection to connect to a Kafka broker over SSL.

You can also specify the configuration properties required to connect to a Kafka broker in a secured way in the Additional Connection Properties or Additional Security Properties in the Kafka connection. You can also specify the configuration properties required to connect to a Kafka broker in a secured way in the Consumer Configuration Properties or Producer Configuration Properties in the Kafka Source or Target transformation.

How can I configure properties specific to a Kafka consumer or producer?

You can use the following mapping properties to configure properties specific to a Kafka consumer or producer:

- Consumer Configuration Properties. Specify configuration properties specific to a Kafka consumer. For more information about consumer configuration properties, see Kafka documentation.
- Producer Configuration Properties. Specify configuration properties specific to a Kafka producer. For more information about producer configuration properties, see Kafka documentation.

How can I configure Consumer Configuration Properties?

You can configure the Consumer Configuration Properties for a Kafka source on the **Mapping** tab. If you read data from a Kafka cluster that is configured for Kerberos authentication, specify the following property to override the **Additional Connection Properties** specified in the Kafka connection:

```
security.protocol=SASL_PLAINTEXT,sasl.kerberos.service.name=kafka,sasl.mechanism=GSSAPI
```

How can I configure Producer Configuration Properties?

You can configure the Producer Configuration Properties for a Kafka target on the **Mapping** tab. If you write data to a Kafka cluster that is configured for Kerberos authentication, include the following property to override the **Additional Connection Properties** specified in the Kafka connection:

```
security.protocol=SASL_PLAINTEXT,sasl.kerberos.service.name=kafka,sasl.mechanism=GSSAPI
```

You can provide the **Batch Flush Time in milliseconds** and **Batch Flush Size in bytes** properties in the producer configuration properties.

For example,

```
batch.size=100,linger.ms=5
```

What are the message recovery strategies supported for a Kafka mapping?

When you configure a mapping task, you can configure the **Recovery Strategy** property in the **Advanced Session Properties** section on the **Schedule** tab of the mapping task and select **Resume from the last checkpoint**. The Secure Agent saves the mapping state of operation and maintains target recovery tables. If the mapping aborts, stops, or terminates, the Secure Agent uses the saved recovery information to resume the mapping from the point of interruption.

How can I terminate a Kafka mapping?

As Kafka Connector processes the Kafka messages in batches, the Secure Agent terminates all the mappings automatically.

What are the data sources and targets that can be supported by Kafka Connector?

Kafka Connector supports data sources and targets supported by Cloud Data Integration.

Does Kafka Connector support File Based Recovery Mechanism?

Yes, Kafka Connector supports File Based Recovery when you write data to Kafka topics.

Which Kafka data types does Kafka Connector support?

For information about the various data types that Kafka Connector supports, see the chapter "Kafka Data Type Reference" in the *Cloud Data Integration Kafka Connector documentation*.

Author

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